Avian Knowledge Network Data Sharing Levels (ver 4)

Michael Fitzgibbon
April 19, 2021

The Avian Knowledge Network works to leverage data, information, and partnerships to revolutionize avian conservation. In order to leverage data, we must describe and define clear, distinct sharing levels that express how data collectors and owners want to share their data with others. Our goal is to encourage Data Owners, or Project Leaders, to share as many avian observations as widely as possible for conservation purposes, while acknowledging the right of each organization to benefit from the intellectual investments associated with the data.

As data are added to our systems, each record is flagged with a Data Sharing Level providing explicit control to data owners for how visible each observation is for accessing, querying, summarizing, visualizing, and analyzing. Data Sharing Levels allow data owners to grant full access without restrictions, require a data sharing agreement for full access, allow data to be included in summary tools, only share metadata, or restrict sharing altogether. Data Sharing Levels are adaptable to the data owner's current needs, allowing data owners to, say, restrict access when observations are first recorded and broaden the availability over time as the sensitivity of the data diminishes.

Project Leaders manage sharing levels per sampling event (a collection of observations made at a sampling unit location at a given date/time) in a project, making all observations within that event consistently shared as a unit, ensuring analytical integrity.

All sampling events are associated with a named sampling unit location, and those spatial coordinates are shared openly by default. In addition to restricting data availability by managing Data Sharing Levels, Data Owners can also hide sampling unit locations from being discoverable. This will hide such locations from applications that provide a geographic view of where observations are located (eg, AKN Observations Map),

The AKN Data Sharing Levels are broken into two stages: *In Review* and *Ready for Use*. During the *In Review* stage, Project Leaders manage their project or organization data workflow for entering and proofing observations. Once the data are reviewed, the *Ready for Use* stage allows the data owner to choose how broadly they want to share their data.

As data move between these stages and the levels within each stage, all AKN applications are responding accordingly, limiting access based on the person's role using the application as well as the specific Data Sharing Level assigned. A Project Leader can change accessevels on data records at any time as they see fit. Note that once data have been shared openly and used outside of the AKN, changing those data to a more restricted level will only control future access.

The *In Review* stage levels are as follows:

***In Review* Level**	**Description**
RAW	Data are freshly entered or uploaded, and are only available to Project members. RAW is the default level for all data added into the AKN.
CLEAN	Data that have undergone a first level of review, often by the person entering or loading the data, and are only available to Project members.
AVAILABLE	Data have been fully reviewed and deemed ready for the data owner to assign a Ready for Use stage Level (see below). AVAILABLE data are only visible to Project data owners.
RESTRICTED	Data have been fully reviewed, but the data owner does not want to share the observations at all. ~~(~~Note that projects always share project names, descriptions, and contact information~~)~~. No detailed metadata, data summaries, nor observation information are ~~is~~ shared outside of the project.

Once data is AVAILABLE, the *Ready for Use* stage levels available are as follows:

***Ready for Use* Level**	**Description**
METADATA ONLY	METADATA ONLY level shares the project description and contact, survey locations, survey methods, and the species surveyed. (Maps to AVAILABLE LEVEL 1 in AKN sharing levels v1)
SUMMARIZE ONLY	SUMMARIZE ONLY level inherits all METADATA ONLY rules, plus allows use of data as part of publically available visualizations and summaries without survey effort and specific count details. (Maps to AVAILABLE LEVEL 2 in AKN sharing levels v1)

SHARE WITH PERMISSION	SHARE WITH PERMISSION level inherits all SUMMARIZE ONLY rules, plus allow use of full observation data and metadata detail to anyone that has a data sharing agreement with the data owner. (Maps to AVAILABLE LEVEL 3 or 4 in AKN sharing levels v1)
SHARE OPENLY	SHARE OPENLY level allows full observation data and metadata detail to anyone for conservation uses without the need for explicit permission from the data owner. (Maps to AVAILABLE LEVEL 5 in AKN sharing levels v1)

It's helpful to see exactly how the AKN applications will respond specifically to all Data Sharing Levels for all user roles:

				Data Sharing Levels							
				In Review stage				Ready for Use stage			
Tool	**Activity**	**Role**	**Data Location**	RAW	CLEAN	AVAILABLE	RESTRICTED	METADATA ONLY	SUMMARIZE ONLY	SHARE WITH PERMISSION	SHARE OPENLY
Bulk Uploader	add new datasets	Project Leader or Biologist	Projects	x							
Biologists	add new visit	Project Leader or Biologist	Projects	x							
Biologists	edit visit / observation data	Project Leader	Projects	x	x	x	x	x	x	x	x
Biologists	edit visit / observation data	Biologist	Projects	x	x						
Biologists	set data sharing levels	Project Leader	Projects	x	x	x	x	x	x	x	x
Project Leaders	add / edit project metadata (locations, methods, people)	Project Leader	Projects	x	x	x	x	x	x	x	x
Project Leaders	download datasets	Project Leader	Projects	x	x	x	x	x	x	x	x
Project Leaders	set data sharing levels	Project Leader	Projects	x	x	x	x	x	x	x	x
Data Sharing Level	set data sharing levels	Project Leader	Projects	x	x	x	x	x	x	x	x
Analysts	analyze observations	Project Leader	Warehouses			x	x	x	x	x	x
Phenology Tool (includes ECOS IPaC)	use	Public	Warehouses						x	x	x
Observations Map and node specific maps	use	Public	Warehouses						x	x	x
Data Downloader	find and download observations	Project Leader	Warehouses					x**	x**	x**	x
Data Downloader	find and download observations	Public	Warehouses								x

x** = available only for projects they are Project Leader on.

Roles listed above:

- Biologist - a registered user who can add and edit observation data in a project.
- Project Leader - a registered user who is the Data Owner, or acts on behalf of the Data Owner, for one or more projects who can add and edit project metadata and observation data, and set Data Sharing Levels for their project(s).
- Program Leader - a registered user who is Project Leader for a Program.
- Public - an unregistered user.

Data Locations listed above:

- Projects - data managed and controlled by the assigned Project Leaders, supporting live editing, updating, and curation of metadata and observation data. Data in projects are organized to segregate user access and support organizational change control, and allowing for a mixture of protocol types.
- Programs - a collection of Projects that have Program Leaders who act as Project Leaders on all of the Projects in the collection.
- Warehouses - project data automatically copied from Projects by the system nightly to support conservation application, including summaries, analysis, and visualization. Data in warehouses are organized by protocol types, supporting a mixture of projects.

One application that is not listed in this table is the Data Catalog, which supports an independent database that provides discoverable information about projects. A Project Leader can decide how much information to include or not in their Data Catalog records. Changing Data Sharing Level information on observation data has no impact on information available in the Data Catalog.

Avian Knowledge Network Data Sharing Levels (ver 3)

Michael Fitzgibbon
May 29, 2019

The Avian Knowledge Network works to leverage data, information, and partnerships to revolutionize avian conservation. In order to leverage data, we must understand how data collectors and owners want to share their data with others. We encourage data owners to share as many avian observations as widely as possible for conservation with the understanding that each organization's needs are different.

As data is added to our systems, each record is flagged with a Data Sharing Level providing explicit control to data owners for how visible each observation is for querying, summarizing, visualizing, and analyzing without explicit permission from the data owner. Data Sharing Levels are adjustable, allow data owners to, say, restrict access to start and broaden the availability over time as the sensitivity of the data diminishes.

Users manage sharing levels per observation visit, making all observations within a visit consistently shared as a unit, ensuring analytical integrity.

The AKN Data Sharing Levels are defined as follows, with each level broadening the scope of sharing as you proceed down the list:

Sharing Level	Brief Desc	Full Description
RAW	New	A pre-sharing level, RAW data are new data entered or uploaded, and are only available to Project members. RAW is the default level for all data added into the AKN.
CLEAN	Proofed	A pre-sharing level, CLEAN data have undergone a first level of review, often by the person entering or loading the data, and are only available to Project members.
AVAILABLE	Analysis-ready	A pre-sharing level, AVAILABLE data have been fully reviewed and deemed ready for analysis, but only to Project data owners.

RESTRICTED	Not shared	RESTRICTED data only share the project name, description, and contact. No metadata, data summaries, nor observation information is shared outside of the project. This Level clearly indicates your need to not share data.
LEVEL 1	Metadata shared	LEVEL 1 data only share the project description and contact, survey locations, survey methods, and the species surveyed.
LEVEL 2	Summarized	LEVEL 2 data are shared through publically available visualizations and summaries that obfuscate observation details.
LEVEL 3/4	Shared with permission	LEVEL 3 and 4 data are shared with full observation detail to anyone that has a data sharing agreement with the data owner.
LEVEL 5	Shared openly	LEVEL 5 data are freely available for use by anyone for use conservation analysis without explicit permission by the data owner.

These AKN Data Sharing Levels determine impact how your data will be visible within a variety of AKN tools. Here is a list of how each of the AKN tools respond to Data Sharing Levels:

AKN Tool	User Roles	Data Sharing Levels (DSL)
Biologists	Project Leaders (data owner) and Biologists (data enterers) have access by Project.	As data for a visit is first entered by any Project member, the DSLis RAW. After they've reviewed and corrected their visit observation, the DSL is CLEAN. Project Leaders can change the DSL on any visit to any DSL. Once the Project Leader changes the DSL beyond RAW or CLEAN, Biologists on the project can no longer edit the visit.
Project Leaders	Project Leaders only have access by Project	The DSL can be changed on any visit. All data is available to the Project Leader regardless of DSL.
Bulk Uploader	Biologists and Project Leaders have access	All data bulk uploaded is set to RAW.

	by Project	
Data Sharing Level	Project Leaders only have access by Project	All visits can have DLS changed in bulk.
Analyst	Project Leaders only	Any data that is not RAW can be analyzed. We allow users to analyze CLEAN data as a quality assurance technique.
Data Catalog	Public, Project Leaders	All projects are listed in the Data Catalog, with titles, descriptions, and contact information. Any project metadata available for observations at DLS LEVEL 1 or higher will be included in the description.
Data Downloader	Public, Project Leaders	Any Project Leader can download any data for their Projects at DLS LEVEL 3 or higher. Anyone else can download any data at DSL LEVEL 5.
Phenology Tool and IPaC	Public	Includes any data at DSL LEVEL 2 or higher (models rerun periodically).
Observations Map and node specific maps	Public	Includes any data at DSL LEVEL 2 or higher.

AKN Data Sharing Levels

Michael Fitzgibbon
November 28, 2018

Note: I have added a candidate Data Sharing Levels v3 at the end of this document.

The following outlines suggested changes to how we describe AKN Data Sharing Levels. The *Proposed Language* section reflects changes as it would be described most fully on the AKN website and other similar venues. The two *Current Language* sections have the way we describe the sharing levels both on the AKN website, as well as in field help within our web apps. We will want an abbreviated version of the *Proposed Language* written for the applications after we've settled on the fuller text.

Please make edits in Suggesting mode so I can see what people are changing.

Proposed Language

Avian Knowledge Network's data sharing levels are how data owners mark their data records and communicate their intention of how they want data made available, discoverable, and shareable with partners. The AKN has five different sharing levels that provide increasing levels of visibility and use.

The AKN data warehouses, organized for query, reporting and analysis, are where the data sharing levels are utilized. As data owners mark their data in the project databases, these records are published in the warehouses for use by a wide variety of tools. Each tool is aware of the access restrictions set by these sharing levels, and uses only those data that meet the minimum level required for its purpose.

The data sharing levels for the AKN are defined as follows:

Level 1 - Metadata about the datasets, including survey locations, survey methods, and data owner points of contact, are made available to any application or service. Observation data summaries are not available. This level is appropriate for data owners who do not want observations included in summary analytics, and want requests for copies of datasets to come directly to them, not the AKN. Applications, such as the Data Catalog

(https://data.pointblue.org/apps/data_catalog/) or metadata services like DataOne can publish these metadata for anything shared at this level or higher.

Level 2 - All characteristics of Level 1, plus the observations can be summarized for certain publicly available, predefined visualizations (i.e. maps and graphs), but direct access to the data is prohibited. Observation data ~~is~~ are not available from the AKN. This level is appropriate for data owners who are okay with their observations being included in summary analytics automatically, but want requests for copies of datasets to come directly to them, not the AKN. Current applications which summarize data at this level include the Phenology Tool (https://data.pointblue.org/partners/natnode/index.php/phenology-tool/) and Observation Map (https://data.pointblue.org/partners/natnode/index.php/observations-map/) available at various AKN Node Data Center websites.

Level 3 - All characteristics of Level 2, plus the complete data set is available through the AKN upon request, subject to approval from the original data provider through a data sharing agreement. This level is appropriate for data owners who want their observations ~~included in summary analytics automatically, and want their datasets to be~~ shared by the AKN, ~~so long as a data sharing agreement is in place first~~.

Level 4 - All characteristics of Level 3, plus primary occurrence data (species, location, and date- no counts and no survey effort information) can be made available to existing bioinformatic efforts (such as GBIF) without needing a request. This level is appropriate for data owners who want their observations included in non-AKN summary analytics and bioinformatic efforts automatically, and want their datasets to come from the AKN, ~~so long as a data sharing agreement is in place first~~.

Level 5 - Data ~~is~~ are freely available, including all occurrence data details, through all tools of AKN without prior data sharing agreements. This level is appropriate for data owners who want to share data widely and freely amongst partners of the AKN. All applications that use the AKN data warehouses, such as the Data Downloader tool (https://data.pointblue.org/apps/downloader/), will have these data freely usable by anyone accessing these tools.

In addition to data sharing levels, the AKN has additional levels included that are restricted from sharing, but help data owners manage the entire data lifecycle. Their extended levels include:

Raw - Data were input into project databases, but no further review or processing has taken place. Data are available for project use only and not available to the AKN. Data entry applications, such as Biologists (https://data.pointblue.org/science/biologists/), will have data

in this level after they are first entered and saved by the user.

Clean - Data were reviewed in project databases by member(s) of the project team. Data are available for project use only and not available to the AKN. Data entry applications, such as Biologists (https://data.pointblue.org/science/biologists/), will have data in this level after a user has reviewed data entered and marked as "Proofing completed".

Approved - Data were reviewed by project management, but no indication has been made of AKN data sharing levels. Data are available for project use only and not available to the AKN. Approved data are available to the data owners through the Analysts tool for reporting, visualization and further inspection.

Restricted - Data and metadata are not available to the AKN nor shared in any form with AKN partners automatically. All queries and access to data must come through requests to the contributing data owners. This level is appropriate for data owners who do not want their data available in any form to the AKN partners, with all requests for metadata or observation records to come directly to them.

Current Language - field help in Biologists

(note: we should have same language in "Set Data Sharing Levels" application, but currently data sharing levels aren't explained there.)

The following is a list of available Data Sharing levels:

Status	Description
RAW	Raw Input − Unreviewed − do not use outside of project
CLEAN	Biologist Reviewed − awaiting project leader approval − do not use outside of project
RESTRICTED	Restricted access − clean and project leader verified data, but must contact Point Blue Conservation Science or contributing institution project leader or division director before using. Data is not distributed to other Avian Knowledge Network (AKN) partners automatically.
APPROVED	Available for analysis − internal Point Blue Conservation Science or contributing institution use only

AVAILABLE Level 1	Same as APPROVED but also shared with AKN: Data are stored in the AKN's primary data warehouse. The warehouse serves as the primary archive of all AKN data, and no applications connect directly to the warehouse. Instead, data from the warehouse are ported to separate data views created specifically to optimize the performance of an application that connects to it. Data owners can specify how their data can be used in the data views, with the option that their data are not exposed to the public at all.
AVAILABLE Level 2	Same as AVAILABLE Level 1 with the following addition: data can be used in certain publicly available, predefined visualizations (i.e. maps and graphs), but direct access to the data is restricted.
AVAILABLE Level 3	Data are used in publicly available, predefined visualizations (i.e. maps and graphs). Additionally, the complete BMDE data set is available upon request, subject to approval from the original data provider.
AVAILABLE Level 4	Data can be used in publicly available, predefined visualizations (i.e. maps and graphs) and also may be available upon request. Additionally, some components of the data are made available to existing bioinformatic efforts (GBIF and ORNIS). These bioinformatic efforts only provide the data "marked-up" to Darwin Core, used to describe primary occurrence (location, date and species for example).
AVAILABLE Level 5	Data are used in publicly available, predefined visualizations (i.e. maps and graphs), and are available to existing bioinformatic efforts. Additionally, the complete BMDE data set is available for download directly via download tools.

Current Language - AKN website

The following are the Avian Knowledge Network's data sharing levels. These are applicable to each and every record in the network individually, so that different records may have different access levels. Data published under one of the five Levels below are stored in the AKN's primary data warehouses. The warehouses serve as the primary archives of all AKN data. No applications connect directly to the warehouses, but data from a warehouse are ported to separate data views created specifically to optimize the performance of an application that connects to it. Data owners can specify how their data can be used in the data views, with the option that their data are not exposed to the public at all.

Level 1 - Some information is made available to others than project members about the data. Specifically, only metadata about the datasets are made available to any application or service.

Level 2 - Same as Level 1 with the following addition: data can be used in certain publicly available, predefined visualizations (i.e. maps and graphs), but direct access to the data is restricted.

Level 3 - Data are used in publicly available, predefined visualizations (i.e. maps and graphs). Additionally, the complete BMDE data set is available upon request, subject to approval from the original data provider.

Level 4 - Data can be used in publicly available, predefined visualizations (i.e. maps and graphs) and also may be available upon request. Additionally, some components of the data are made available to existing bioinformatic efforts (GBIF and ORNIS). These bioinformatic efforts only provide the data "marked-up" to Darwin Core, used to describe primary occurrence (location, date and species for example).

Level 5 - Data are used in publicly available, predefined visualizations (i.e. maps and graphs) and are available to existing bioinformatic efforts. Additionally, the complete BMDE data set is available for download directly via download tools.

Some nodes have extended these levels to include additional levels to help users manage the entire data lifecycle. Their extended levels include:

Raw - Data were input but no further review or processing has taken place. Data are available for project use only and not to the AKN.

Clean - Data were input and reviewed by member(s) of the project team. Data are available for project use only and not to the AKN.

Approved - Data were reviewed by project management, but no indication has been made of AKN data sharing levels. Data are available for project use only and not to the AKN.

Restricted - Same as APPROVED and not distributed and shared to other AKN partners automatically. All access to data must come through requests to the contributing institution project management.

11/20/2018 - pasted from Nick's 11/20/2018 email to the AKN SC

	Data Sharing Levels				
Information Available	**1**	**2**	**3**	**4**	**5**
Metadata	A	A	A	A	A
Survey Protocol	A	A	A	A	A
Data Owner POC	A	A	A	A	A
Discoverable via Data Catalogue	A	A	A	A	A
Data Summaries	RO	A	A	A	A
Primary Occurrence Data	RO	RO	RA	A	A
Raw Data Records	RO	RO	RA	RA	A
Survey Locations	RO	RO	RA	RA	A

A = Freely Available

RO = Data access provided by data owner

RA = Data access provided by AKN

Data Sharing Levels v2 proposal

This proposal names the sharing levels (the numbers are hard to remember and are confusing), simplifying their distinctions, and reduces the number of levels by one. In parentheses are how existing v1 sharing levels would map into these.

Raw -- fresh entry, only available to project members
Clean -- reviewed, only available to project members
Available -- fully reviewed, only available to project members
Hidden (RESTRICTED) -- no information shared outside of project
Metadata only (LEVEL 1) -- only project description, summary metadata, and contact information shared
Summarize only (LEVEL 2) -- contact information available, used in summaries
Shared with permission (LEVEL 3 + 4) -- data shared, use restricted by needing a data sharing agreement
Open (LEVEL 5) -- data share, open use

Hidden	Metadata	Summary	Shared	Open	
X					Project name, description, and contact information only available.
	X	X	X	X	Metadata available, including contact information, locations, survey methods, species surveyed
		X	X	X	Used in Online summaries that appropriately obfuscate data (map, phenology, others)
			X	X	Shared / available for download, requires getting data sharing agreement
				X	Shared / available for download, open use without permission

Notes:

- For “Shared / available for download”: the data set can be downloaded with user registration, letting people know that all data marked as “Shared with permission” cannot be used without a data sharing agreement. We should make this clear in a popup dialog before download, and a README provided in the download packages

about this. I suggest we don't have the systems police Level 3 downloads. Data owners should be able to discover who has downloaded any "Shared with permission" or "Open" data. Anyone discovered downloading data and not getting permission to use could have access turned off in our systems.

- Hiding any metadata, including locations, is not managed through sharing levels. Sampling Units should be able to be hidden entirely, obfuscated, or shared openly in Project Leaders. All original locations would be used in online summaries. All data shared that is obfuscated would be noted in the download (column of locational accuracy or obfuscation indication added, with notes in README)

MEGHAN S: Here's my attempt at a re-write. I still have a bunch of questions. Can someone help me answer these? Thanks!

The following are the Avian Knowledge Network's Data Management Levels and Data Sharing Levels. These are applicable to each and every record in the network individually, so that different records may have different management and sharing levels. Data Management Levels are separate from Data Sharing Levels and simply indicate the level of review a record has undergone by the data owner (s). A record can be assigned a Data Sharing Level no matter what Data Management Level it has been assigned **(is this right? Or does it have to be "Available" before being given a sharing level?)**. Data published under one of the five Data Sharing Levels below are stored in the AKN's primary data warehouses. The warehouses serve as the primary archives of all AKN data. No applications (tools) connect directly to the warehouses, but data from a warehouse are ported to separate data views created specifically to optimize the performance of the individual tool (i.e. Data Catalog, Data Downloader, Phenology Tool, Observations Map and node-specific mapping tools or IPaC) that uses it. Data owners can specify how their data can be used in the data views, with the option that their data are fully hidden/restricted (not exposed at all). To learn about how to assign data access levels to a record, view the **video tutorial on this topic**.

Data Management Level
Assigns a level of review to an individual record

Raw - Fresh entry, only available within a project
Clean - Reviewed, only available within a project
Available - Fully reviewed, only available within a project

Data Sharing Level

Designates what tools can use a particular record and what level of access the general public can have to information about that record and its associated project.

RESTRICTED - Metadata at the project level only is available in the Data Catalog, including contact information.

LEVEL 1 - Metadata available in the Data Catalog, including contact information, locations, survey protocols, species surveyed.

LEVEL 2-4 - Metadata available in the Data Catalog, including contact information, locations, survey protocols, species surveyed. Used in online publicly available tools that summarize data (Observations Map and node-specific mapping tools, Phenology Tool, and IPaC).

LEVEL 5 - Metadata available in the Data Catalog, including contact information, locations, survey protocols, species surveyed. Used in online publicly available tools that summarize data (Observations Map and node-specific mapping tools, Phenology Tool, and IPaC). Subset of data fields available for download via the Data Downloader without permission.